Scuola universitaria professionale della Svizzera italiana
Dipartimento tecnologie innovative

SUPSI

Minesweeper Project

Studente/i

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Bechelor in Computer Science

Modulo / Codice Progetto

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Data

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Context and Motivation

- Development of a desktop game (Minesweeper)
- Application of principles of Software Engineering
- Use of 2 modules:
  - Frontend: use of MVC architectural pattern
  - Backend: use of 3-layer architecture
- Use of JavaFX for the interaction with the user
- Submission for Software Engineering Evaluation

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Problem

Requirements:

- Implements MVC architecture
- Interactive GUI using JavaFX
- UML diagram
- internazionalization
- Backend / Frontend modules
- User preferences and interaction
- Multiplatform

Constraints:

- Programming Language: Java
- Dependency Management: Maven
- Versioning Management: Git
- Issue Management: Gitlab
- Automatic build pipeline
- Time Constraints

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State of the Arts

- **Review of existing Minesweeper implementations**
- **Brief summary highlighting standard functionalities and features**
- **Minesweeper:** https://www.pragmaticcoding.ca/javafx/minesweeper
- **Project Structure:** https://www.baeldung.com/

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Waterfall / Agile approach

- **Hybrid Development Approach**
- **Task Decomposition**
- **Immediate Testing**
- **Continuous Integration**
- **Incremental Delivery**

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Approach

- Use of 2 modules based of the dependency with the UI
- Use of JavaFX in frontend
- Use of i18n, l10n in backend
- UML diagrams:
    - Use Cases: define user interactions and goals
    - State machine diagram: shows game states and transitions
    - Activity diagram: outlines workflow steps (e.g., “Save Game”)

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Frontend approach

Managing of MVC:

- **Model: connect the frontend with the backend**
- **View: user interface (JavaFX):**
    - **Button**
    - **ComboBox**
    - **Spinner**
- **Controller: user interaction, updates the Model and View**

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Backend approach

Managing of 3-layer application:

- **Application: connect the backend to the frontend**
- **Business: game logic (cell states, bomb placement, Minesweeper rules)**
- **Data: handles persistence (saving/loading game state and settings to/from JSON files)**

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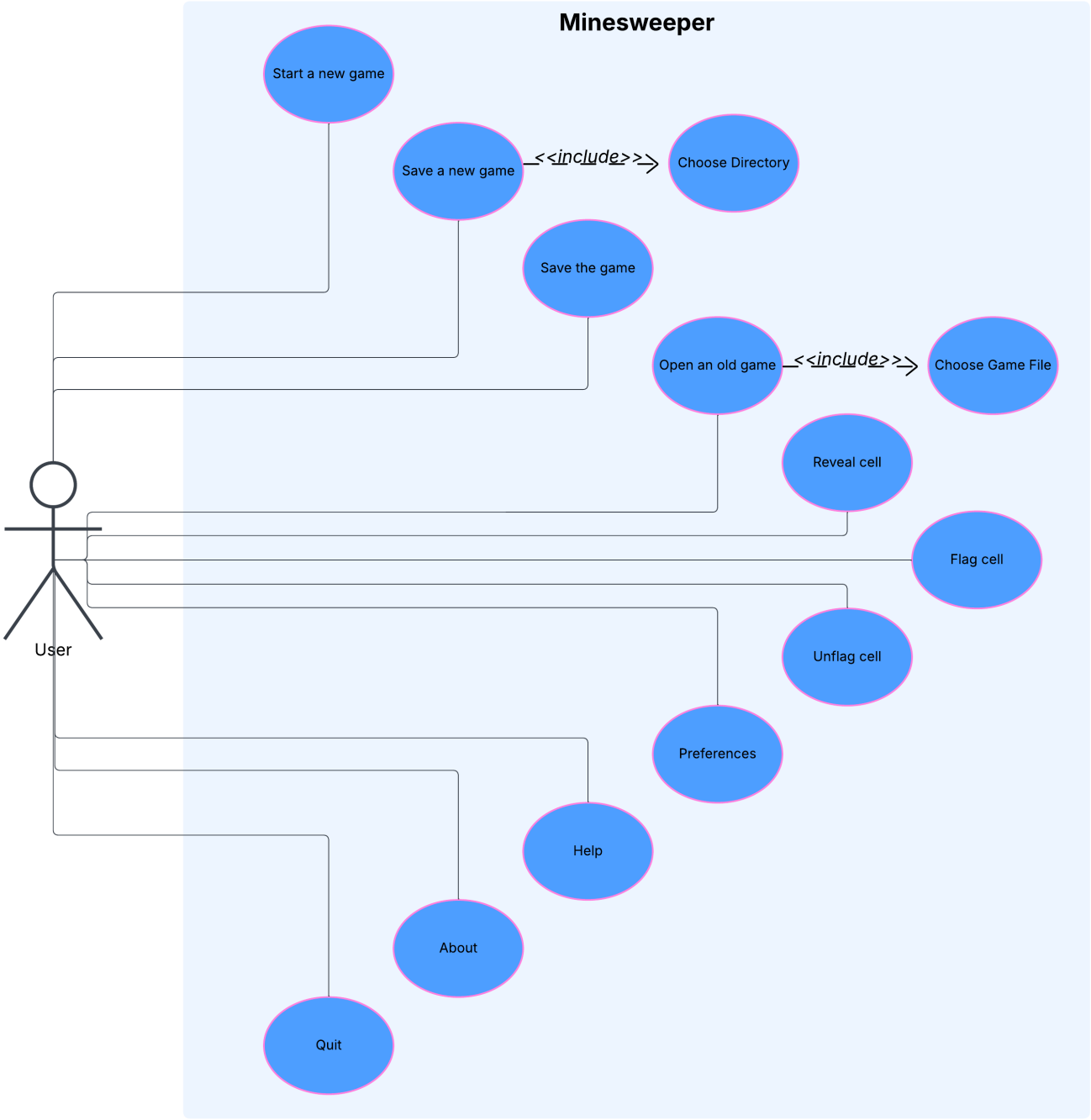
Use cases



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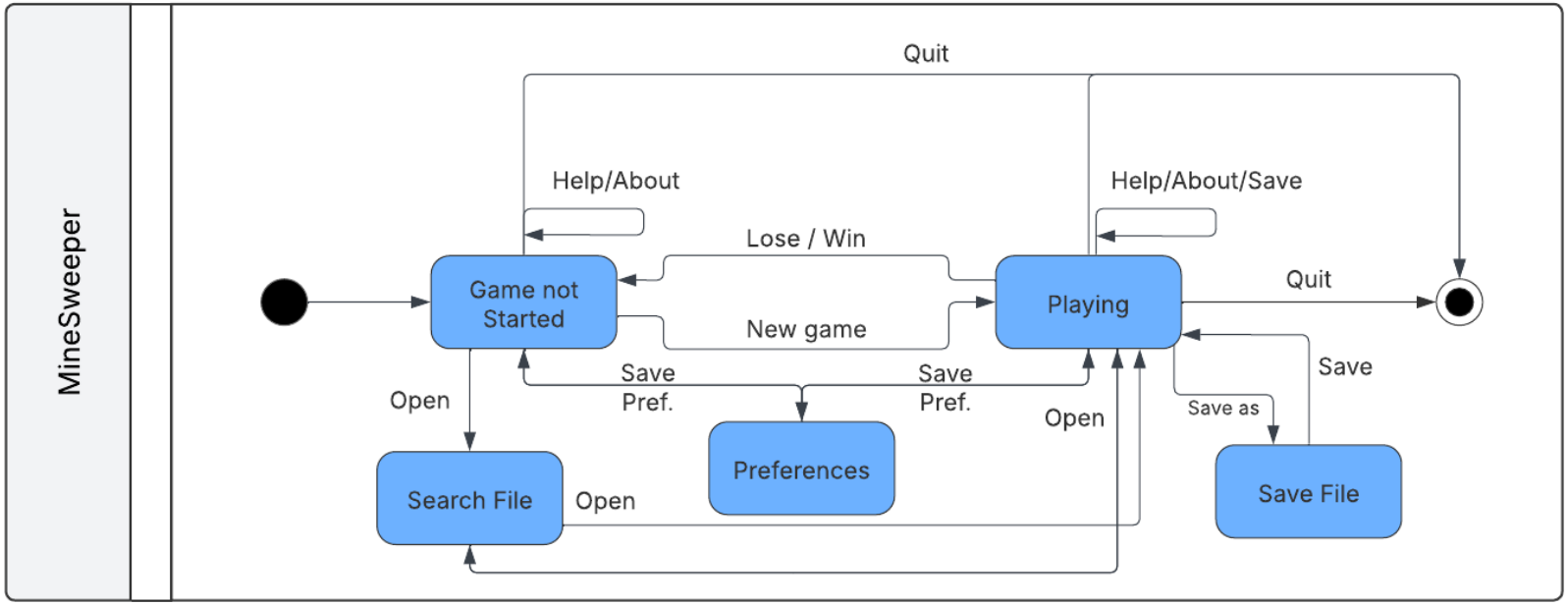
State machine diagram



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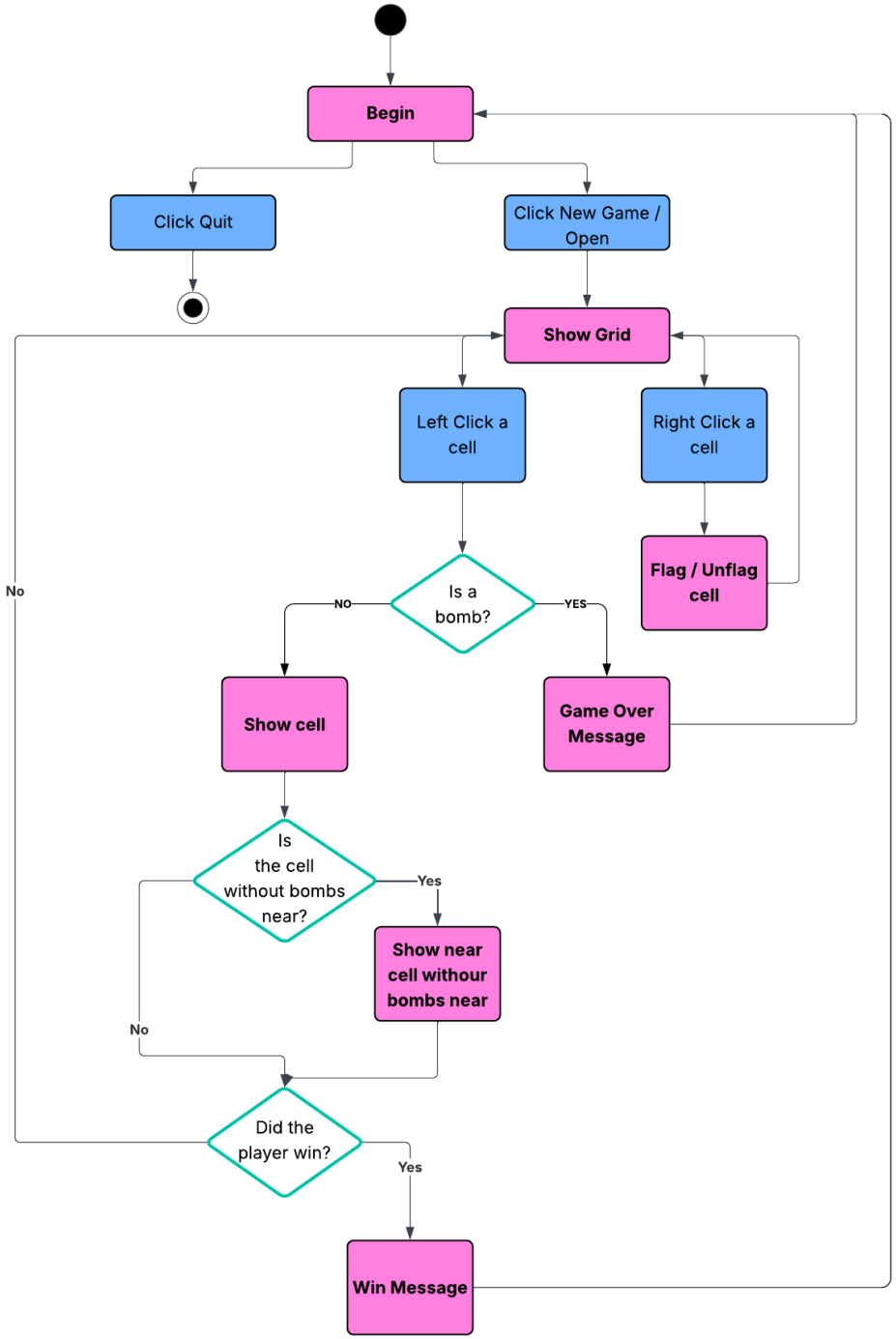
Activity diagram



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Result

- Play a new game
- Save a started game
- Open an old game
- Change the number of bomb
- Change the language (IT/EN)

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Future implementation

- **Leaderboard & Stats Tracking**
- **Intelligent Hints & Solver**
- **Configurable Grid Size**
- **Analytics & Achievements**

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Demo

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Thank you for the attention

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